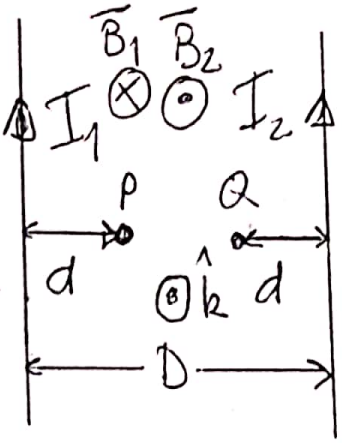
1) a)



$d = 5\,cm$

$D = 20\,cm$

$0 = B_P \rightarrow 0 = \frac{\mu_0 I_1}{2\pi d} - \frac{\mu_0 I_2}{2\pi(D-d)}$

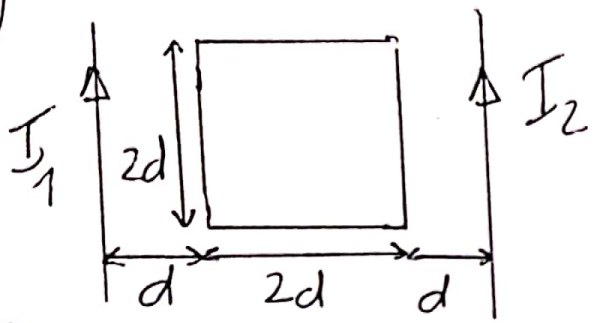
$\rightarrow \frac{I_1}{d} = \frac{I_2}{D-d} \rightarrow \boxed{I_2 = 3I_1 = 6A}$

(SENTIDOS IGUALES)

$\vec{B}_Q = \frac{\mu_0}{2\pi}\left(-\frac{I_1}{D-d} + \frac{I_2}{d}\right)\hat{k} = \frac{\mu_0 I_1}{2\pi}\left(-\frac{1}{D-d} + \frac{3}{d}\right)\hat{k}$

$\rightarrow \boxed{\vec{B}_Q = 21{,}33\,\mu T\,\hat{k}}$

b)



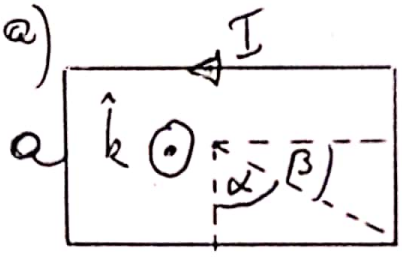
$\phi = \phi_2 - \phi_1 = \frac{2\mu_0 d}{2\pi}\left[I_2 \ln\left(\frac{3d}{d}\right) - I_1 \ln\left(\frac{3d}{d}\right)\right]$

$\phi = \frac{2\mu_0 d}{\pi}\ln(3) I_1 \rightarrow \boxed{\phi = 8{,}79 \times 10^{-8}\,T\,m^2}$

c) $\phi_1 = \frac{2\mu_0 d}{2\pi}\ln(3) I_1$, $\mathcal{E} = \frac{2\mu_0 d}{2\pi}\ln(3)\,100\pi\,\frac{1}{s}\,2A\cos\left(100\pi\,\frac{1}{s}t\right)$

$\rightarrow \mathcal{E} = 13{,}8\,\mu V \cos\left(100\pi\,\frac{1}{s}t\right)$, EN $t = 0{,}01\,s \rightarrow \boxed{|\mathcal{E}| = 13{,}8\,\mu V}$

2) a)



$tg\alpha = 2 \rightarrow \alpha = 63{,}435°$; $\beta = 26{,}565°$

$B = 2\frac{\mu_0 I}{4\pi a/2}\,2\sin(63{,}435°) + 2\frac{\mu_0 I}{4\pi a}\,2\sin(26{,}565°)$

$\rightarrow \boxed{\vec{B} = 35{,}777\,\mu T\,\hat{k}}$

b) $\frac{\mu_0 \tilde{I}}{2R} = B_0 \rightarrow \boxed{R = 9{,}88\,cm}$

3) $p = v_{\parallel} T = v_{\parallel}\frac{2\pi m}{|q|B_1} = \frac{2\pi m v}{|q|B_1}\cos 80° \rightarrow \frac{mv}{|q|} = \frac{pB_1}{2\pi\cos 80°}$

EN EL CASO $\alpha = 90°$:

$R = \frac{mv}{|q|B_2} = \frac{pB_1}{2\pi B_2 \cos 80°} \rightarrow \boxed{R = 1{,}04\,mm}$

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4) a) $V_{ef} I_{ef} = S \rightarrow V_{ef} \cdot 0{,}6A = 60\,VA \rightarrow V_{ef} = 100\,V$

$Z = \frac{V_{ef}}{I_{ef}} = \frac{100\,V}{0{,}6A} \rightarrow Z = 166{,}67\,\Omega$

$\cos\phi = \frac{R}{Z} \rightarrow R = 166{,}67\,\Omega \cdot \cos 30° \rightarrow \boxed{R = 144{,}34\,\Omega}$

b) $\omega_R^2 = \frac{1}{LC} \rightarrow \left(100\pi \frac{1}{s}\right)^2 = \frac{1}{LC}$

$(X_L - X_C)^2 = Z^2 - R^2 = 6944{,}85\,\Omega^2 \rightarrow |X_L - X_C| = 83{,}34\,\Omega$, y $X_L < X_C$

$\rightarrow X_C - X_L = 83{,}34\,\Omega \rightarrow \frac{1}{80\pi \frac{1}{s} C} - 80\pi \frac{1}{s} L = 83{,}34\,\Omega \rightarrow$

$\rightarrow \frac{\left(100\pi \frac{1}{s}\right)^2}{80\pi \frac{1}{s}} L - 80\pi \frac{1}{s} L = 83{,}34\,\Omega \rightarrow \boxed{L = 0{,}59\,H;\ C = 17{,}2\,\mu F}$

c) $V_R = R I_{ef} \rightarrow \boxed{V_R = 86{,}6\,V}$

$X_L = 80\pi \frac{1}{s} \times 0{,}59\,H = 148{,}28\,\Omega$; $X_C = \frac{1}{80\pi \frac{1}{s} \times 17{,}2 \times 10^{-6} F} = 231{,}33\,\Omega$

$V_L = X_L I_{ef} \rightarrow \boxed{V_L = 89\,V}$; $V_C = X_C I_{ef} \rightarrow \boxed{V_C = 138{,}8\,V}$

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